VTU PREVIOUS QUESTION PAPER JUL/AUG 2022

QUESTION PAPER INDEX LIBRARY SECTION

M.C.A.

M.CA I
/ 20 I Sem MCA

CBCS SCHEME

USN

20MCA11

First Semester MCA Degree Examination, July/August 2022
Data Structures with Algorithms

Time: 3 hrs. Max. Marks: 100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

Module-1

1 a. What is data structure? Explain the classification of data structure with example. **(06 Marks)**

b. Write the algorithm to convert infix to postfix expression. **(06 Marks)**

c. Write a C program to perform push( ) and pop( ) operation in a stack using arrays. **(08 Marks)**

OR

2 a. What is polish and reverse polish expression? Write an algorithm to convert infix expression to prefix expression. **(07 Marks)**

b. Write a C program to evaluate the postfix expression. **(07 Marks)**

c. Convert the following infix expression to postfix expression:

i) $(A + B * C) / (D \$ P - E * F)$

ii) $A * B / C \$ P + D * E \$ P / G$ **(06 Marks)**

Module-2

3 a. What is recursion? Write a function to implement tower of Hanoi problem with recursion. **(06 Marks)**

b. Write a program to implement double ended queue using array. **(08 Marks)**

c. What is priority queue? Write an algorithm to insert the element in priority queue based on priority. **(06 Marks)**

OR

4 a. What is queue? Write the applications of queue. Write the algorithm to insert and delete an item from queue. **(08 Marks)**

b. Write a program to implement circular queue. **(08 Marks)**

c. What is the advantages of circular queue over general queue? **(04 Marks)**

Module-3

5 a. Explain different types of memory management functions with example. **(10 Marks)**

b. Write a program to insert and delete a node from beginning and end of the link list. **(10 Marks)**

OR

6 a. Explain getnode( ) and freenode( ) operations. **(06 Marks)**

b. Write an algorithm to delete a node from the middle of the linked list. **(08 Marks)**

c. Discuss about static and dynamic memory allocation. **(06 Marks)**

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

Module-4

7 a. Discuss asymptotic notations and basic efficiency classes with example. **(10 Marks)**

b. Explain the various stages of the algorithm design and analysis process with the help of a flowchart. What is the efficiency of algorithm? **(10 Marks)**

OR

8 a. What is an algorithm? What are the characteristics of a good algorithm? Explain with example. Also explain algorithm correctness. **(10 Marks)**

b. Explain the methods to analyze recursive and non-recursive algorithms with example. **(10 Marks)**

Module-5

9 a. Write the algorithm for selection sort and derive the time complexity. **(08 Marks)**

b. Explain Prim's and Kruskal's algorithm, with example. **(12 Marks)**

OR

10 a. Write a function for quick sort. Also analyze its average case time efficiency using recurrence relations. **(08 Marks)**

b. Compare divide and conquer method and greedy method of solving problems with suitable examples. **(08 Marks)**

c. Explain how topological sort algorithm is different from other sorting algorithms. **(04 Marks)**

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First Semester MCA Degree Examination, July/August 2022
Operating System with UNIX

Time: 3 hrs. Max. Marks:100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

Module-1

1 a. Define operating system. Different single processor and multiprocessor in detail. **(10 Marks)**
b. What is system call? Explain the different types of system calls. **(10 Marks)**

OR

2 a. Illustrate the different scheduling algorithms with examples. **(10 Marks)**
b. Explain the three classical problems of synchronization. **(10 Marks)**

Module-2

3 a. What is a deadlock? What are the necessary conditions for deadlocks? **(06 Marks)**
b. Explain the Banker's algorithm with suitable examples. **(10 Marks)**
c. Illustrate the resource-allocation graph. **(04 Marks)**

OR

4 a. Explain the concepts of contiguous memory allocation. **(10 Marks)**
b. For a given reference string calculate the number of page faults using :
i) Optimal Page Replacement ii) Least Recently Used
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1. **(10 Marks)**

Module-3

5 a. What is a file? Explain absolute and relative path names. **(10 Marks)**
b. Briefly explain the Unix file system. **(10 Marks)**

OR

6 a. Illustrate the three standard files. **(10 Marks)**
b. Differentiate between hard links and symbolic links with an example. **(10 Marks)**

Module-4

7 a. What is a process? Explain the mechanism of process creation. **(10 Marks)**
b. Briefly explain the difference between internal and external commands. **(06 Marks)**
c. Write the details of killing processes with singles. **(04 Marks)**

OR

8 a. With syntax and example, explain CASE conditional statement. **(10 Marks)**
b. Illustrate different operations of test command. **(10 Marks)**

Module-5

9 a. What is awk? Explain simple awk filtering with splitting a lines in to fields. **(10 Marks)**
b. Describe : i) Begin and End sections ii)Variable and Expressions. **(10 Marks)**

OR

10 a. Write short notes with respect to : i) Arrays ii) Functions. **(10 Marks)**
b. Write an awk script to compute gross salary of an employee accordingly to a given rule.
HRA – 15%, DA – 45% of basic salary. **(10 Marks)**

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20MCA13

First Semester MCA Degree Examination, July/August 2022
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1 a. Explain OSI network architecture with a neat diagram. **(10 Marks)**

b. Describe the network requirements of application programmer, network operator and network designer with an example. **(06 Marks)**

c. Briefly explain applications of computer networks. **(04 Marks)**

OR

2 a. Explain following ideas with respect to network architecture : layering, protocol, encapsulation, multiplexing and demultiplexing. **(10 Marks)**

b. Consider a point-to-point link 4 km in length. At what bandwidth would propagration delay (at a speed of 2×10^8 m/s) equal transmit delay for 100-byte packets? **(04 Marks)**

c. Explain circuit switched and packet switched networks with a neat diagram. **(06 Marks)**

Module-2

3 a. Apply NRZ, NRZI, Manchester encoding to the following bit stream :
0 0 1 0 1 1 1 1 0 1 0 0 0 0 1 0 **(06 Marks)**

b. Briefly explain Byte oriented protocols. **(06 Marks)**

c. Discuss briefly about wifi and Bluetooth. **(08 Marks)**

OR

4 a. Explain the concept of stop and wait protocol with necessary diagrams for different scenario. **(10 Marks)**

b. A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$.
(i) What is the actual bit string transmitted?
(ii) Suppose the third bit from the left is inverted during transmission. How will receiver detect this error? **(10 Marks)**

Module-3

5 a. Define virtual circuit identifier? Explain virtual circuit network and VC table entry for switches with a neat diagram. **(10 Marks)**

b. Define spanning tree algorithm and explain how it works. **(10 Marks)**

OR

6 a. Explain IPV_4 packet headers format with a neat diagram. **(10 Marks)**

b. Briefly discuss about,
(i) ARP
(ii) ICMP
(iii) DHCP
(iv) Border Gateway Protocol **(10 Marks)**

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Module-4

7 a. Explain three-way hand shakes algorithm for connection establishment and termination respectively for TCP with diagram. **(10 Marks)**

b. Explain UDP header format. **(06 Marks)**

c. What are the key characteristics of UDP? **(04 Marks)**

OR

8 a. Explain congestion avoidance mechanisms in brief. **(10 Marks)**

b. Briefly explain the mechanisms used for TCP congestion control. **(10 Marks)**

Module-5

9 a. Write a brief note on Domain Name System (DNS). **(10 Marks)**

b. Explain the following cryptographic building blocks :

(i) Symmetric-key ciphers
(ii) Public-key ciphers
(iii) Authenticators
(iv) Ciphers

(10 Marks)

OR

10 a. What are firewalls? Explain strengths and weaknesses of firewalls. **(10 Marks)**

b. Explain the following protocols :

(i) SMTP
(ii) HTTP
(iii) SNMP
(iv) IMAP

(10 Marks)

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First Semester MCA Degree Examination, July/August 2022
Mathematical Foundation for Computer Applications

Time: 3 hrs. Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. Statistical tables are permitted.

Module-1

1 a. Define a set, power set and a single ton set with an example for each. **(06 Marks)**
b. State and prove Associative Laws of set theory. **(07 Marks)**
c. Find the eigen values and eigen vectors of $A = \begin{pmatrix} -1 & 3 \\ -2 & 4 \end{pmatrix}$. **(07 Marks)**

OR

2 a. Let $A = \{1, 2, 3, 4\}$ $B = \{2, 4, 5, 6\}$. Find : i) $A \cup B$ ii) $A - B$ iii) $B - A$. **(07 Marks)**
b. 30 cars were assembled in a factory. The options available were a ratio AC and white wall tyres. It is known that 15 of the cars have radios, 8 of them have AC and 6 of them have white wall tyres. Three of them have all 3 options. Determine at least how many cars do not have any option at all. **(07 Marks)**
c. State and prove pigeonhole principle. Prove that if 30 dictionaries in a library contain a total of 61, 237 pages then at least one of the dictionaries must have atleast 2045 pages. **(06 Marks)**

Module-2

3 a. Define the following with an example :
i) Conjunction
ii) Disjunction. **(06 Marks)**
b. Prove that for any 3 propositions p, q, r
$[p \to (q \wedge r)] \Leftrightarrow [(p \to q) \wedge (p \to r)]$. **(07 Marks)**
c. Write the converse, inverse and contrapositive of the statement "If 2 is an integer, then 9 is a multiple of 3". **(07 Marks)**

OR

4 a. Let p, q, r be propositions having the truth values 0, 0 and 1 respectively. Find the truth values of i) $(p \vee q) \vee r$ ii) $(p \wedge q) \wedge r$. **(06 Marks)**
b. Give the direct proof and indirect proof of "If n is an odd integer, then n^2 is an odd integer". **(05 Marks)**
c. Let the unwerse be the set of all integers. Consider the following open statements
$p(x) : x > 3$ $q(x) : x + 1$ is even $r(x) : x \leq 0$.
Write down the truth values of i) $p(2)$ ii) $p(3) \vee \sim r(3)$. **(04 Marks)**
d. Test whether the argument is valid. If Sachin hits a century, then he gets a free car.
<u>Sachin hits a century</u>
$\therefore$ Sachin gets a free car **(05 Marks)**

Module-3

5 a. Let :
$A = \{1, 2, 3, 4\}$
$R = \{(1, 1), (1, 2)(2, 1) (2, 2) (3, 4) (4, 3) (3, 3) (4, 4)\}$
a relation on A. Verify that R is an equivalence relation on A. **(06 Marks)**

b. Let $A = \{1, 2, 3, 4\}$ R be the relation on A defined by xRy iff x divides y. Write down R as the set of ordered pairs and draw the digraph of R. **(07 Marks)**

c. Let $M_R = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ $M_S = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ be the matrix relations of the relations R and S on A.

Find :
i) $R \cup S$
ii) $R \cap S$
iii) $\overline{R}$. Given $A = \{a, b, c\}$. **(07 Marks)**

OR

6 a. Define partition of a set and equivalent class with an example. **(10 Marks)**

b. Draw the Hasse diagram representing the positive divisors of 36. **(10 Marks)**

Module-4

7 a. The probability distribution of a finite random variable X is given by the following table :

x	–2	–1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

Find the value of K, mean and variance. **(10 Marks)**

b. When a coin is tossed 4 times, find the probability of getting.
i) Exactly one head
ii) At most 3 heads
iii) At least 2 heads. **(10 Marks)**

OR

8 a. Find the value of c such that

$$f(x) = \begin{cases} \frac{x}{6} + c & 0 \le x \le 3 \\ 0 & \text{elsewhere is a p.d.f. Also find } P(1 \quad x \quad 2) \end{cases}$$ **(06 Marks)**

b. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that the shower will last for :
i) 10 min or more
ii) Less than 10 minutes
iii) Between 10 and 12 minutes. **(07 Marks)**

c. If x is a normal variable with mean 30 and standard deviation 5 find the probability that
i) $26 \le x \le 40$
ii) $x \ge 45$. **(07 Marks)**

Module-5

9 a. Define the following with suitable examples :
 i) Simple graph
 ii) Null graph
 iii) Complete bipartite graph. **(06 Marks)**

b. Explain Konigsberg bridge problem. **(07 Marks)**

c. Check whether the following graphs are isomorphic.

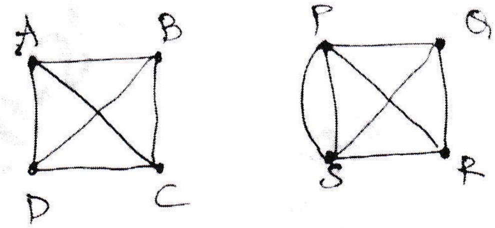

Fig.Q9(c) **(07 Marks)**

OR

10 a. Define the following with an example :
 i) Spanning subgraph
 ii) Induced subgraph
 iii) Planar graphs. **(06 Marks)**

b. Give the graph coloring of the graph shown in Fig.Q10(b) below.

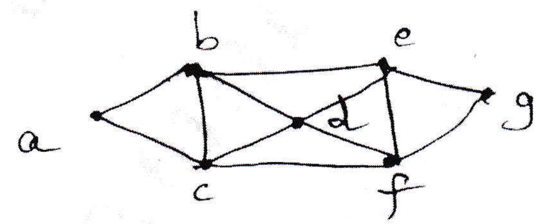

Fig.Q10(b) **(07 Marks)**

c. Use Dijkstra's algorithm to find the shortest path from vertex 1 to each of the other vertices in the weighted directed network shown in the Fig.Q10(c).

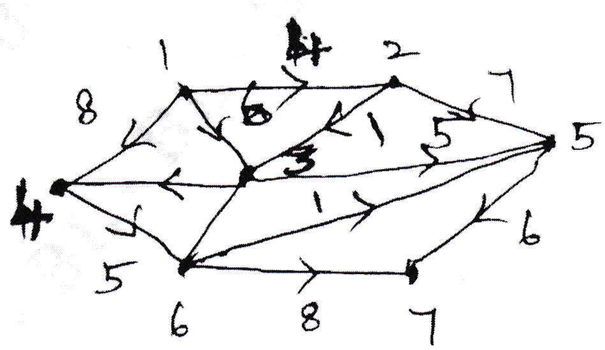

Fig.Q10(c)

Indicate the weight of the shortest paths. **(07 Marks)**

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20MCA21

Second Semester MCA Degree Examination, July/August 2022
Database Management System

Time: 3 hrs. Max. Marks:100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

Module-1

1 a. What is Database Management System? Explain Actors on the scene and Workers behind the scene. **(10 Marks)**

b. Describe the three – Schema architecture of DBMS with a diagram. **(10 Marks)**

OR

2 a. Explain Database system environment with a diagram. **(10 Marks)**

b. Explain centralized and client–server architectures with a diagram. **10 Marks)**

Module-2

3 a. What is schema? Explain Database schema with an example. **(10 Marks)**

b. Explain Primary key and Foreign key with an example. **(10 Marks)**

OR

4 a. What is the Entity? Explain the entity type and Entity sets with a examples. **(10 Marks)**

b. What is the need for normalization? Explain 1NF, 2NF with an example. **(10 Marks)**

Module-3

5 a. What is Datatype? Explain different datatypes in SQL. **(10 Marks)**

b. Explain constraints in SQL with an example. **(10 Marks)**

OR

6 a. Explain aggregate functions in SQL with an example. **(10 Marks)**

b. Explain Group by and having clauses with an example. **(10 Marks)**

Module-4

7 a. What is Trigger? Explain with example. **(10 Marks)**

b. Define views in SQL. Explain with example. **10 Marks)**

OR

8 a. Describe the stored procedure with an example. **(10 Marks)**

b. Explain functions with an example. **(10 Marks)**

Module-5

9 a. Define Transaction. Explain ACID properties with examples. **(10 Marks)**

b. Explain two–phase locking techniques with an example. **(10 Marks)**

OR

10 a. Explain concurrency control based on timestamp ordering with example. **(10 Marks)**

b. What is recovery? Explain recovery in multidatabase systems. **(10 Marks)**

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20MCA22

Second Semester MCA Degree Examination, July/August 2022
Object Oriented Programming with Java

Time: 3 hrs. Max. Marks:100

Note: *Answer any FIVE full questions, choosing ONE full question from each module.*

Module-1

1 a. Explain the Buzzwords of Java. **(10 Marks)**

b. Explain the various primitive data types used in Java. Give suitable examples. **(10 Marks)**

OR

2 a. Define an Array. Write the array declaration syntax for multidimensional arrays. Write a simple Java program to create an array of objects. **(10 Marks)**

b. Demonstrate the concept of garbage collection through suitable examples. **(10 Marks)**

Module-2

3 a. Explain method overloading and constructor overloading with suitable examples. **(10 Marks)**

b. Explain the following :
i) Varargs
ii) Final keyword. **(10 Marks)**

OR

4 a. What are super class constructors? How does super class members are used in Java. **(10 Marks)**

b. What is Method-Overriding? Explain how it allows Java to support run-time polymorphism with an example. **(10 Marks)**

Module-3

5 a. What are interfaces? What are their benefits? Explain how it is implemented in java with suitable example. **(10 Marks)**

b. What is Exception? Explain how exception handling mechanism can be used for debugging a program. **(10 Marks)**

OR

6 a. Define package. Explain the creation of a package using a suitable example program. **(10 Marks)**

b. Develop a simple program and explain multiple catch blocks. **(10 Marks)**

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Module-4

7 a. Define synchronization? Explain how inter-thread communication can be achieved in multithreading using producer and consumer problem. **(10 Marks)**

b. Define multi-threading? Construct a java program to create multiple threads in Java by implementing runnable interface. **(10 Marks)**

OR

8 a. Develop a simple program and explain how enumerations are class types. **(10 Marks)**

b. Analyze the following with syntax and example :
i) values( ) and valuesof( )
ii) compareTo( ). **(10 Marks)**

Module-5

9 a. What are URL classes? Explain URL connections with syntax. **(10 Marks)**

b. What HttpURL connection class. Explain the methods of its. **(10 Marks)**

OR

10 a. Identify different types of methods defined by collection interface. **(10 Marks)**

b. Demonstrate a simple program, the constructors and methods in ArrayList class. **(10 Marks)**

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20MCA23

Second Semester MCA Degree Examination, July/August 2022
WEB Technologies

Time: 3 hrs. Max. Marks:100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

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Module-1

1 a. Write short notes on the following :
i) Web browser
ii) Web server
iii) MIME **(10 Marks)**

b. Discuss the basic structure of a XHTML document. **(05 Marks)**

c. What are the rules to be followed to make use of the html elements in a XHTML document? Explain. **(05 Marks)**

OR

2 a. Demonstrate the use of image maps with a suitable example. **(10 Marks)**

b. What is a hyperlink? Explain the various ways of creating hyper links with example. **(10 Marks)**

Module-2

3 a. Mention any five CSS selectors and explain their use with a suitable example. **(10 Marks)**

b. Discuss on the different ways of including CSS style information to a HTML document. **(10 Marks)**

OR

4 a. What is an Array in Javascript? Explain the various ways of creating arrays. Mention any 5 array methods and explain their use. **(10 Marks)**

b. Write a Javascript program that accepts a value 'n' as input from the user and display the first 'n' fibanocci numbers as output. **(05 Marks)**

c. Write a Javascript program that accepts valid USN as input and displays an error message otherwise. USN format 'XX00 XXX 00' where 'X' is an upper case alphabet, '0' is a digit. **(05 Marks)**

Module-3

5 a. What is Bootstrap? What are the features of Bootstrap? Create a simple web page using Bootstrap. **(10 Marks)**

b. Discuss on the following :
i) Table
ii) Image
iii) Button
iv) Progress bar.
Use Bootstrap code snippet. **(10 Marks)**

OR

6 a. Discuss briefly on the Grid system of Bootstrap. **(10 Marks)**
b. List out the various types of forms in Bootstrap. Explain their use with code snippet. **(10 Marks)**

Module-4

7 a. What is JQuery? What are the advantages of JQuery? **(06 Marks)**
b. Explain the syntax of Jquery script with a suitable example. **(06 Marks)**
c. Write a JQuery program to animate a rectangular Box. **(08 Marks)**

OR

8 a. Mention any ten Jquery selectors and explain their use with code snippet. **(10 Marks)**
b. Demonstrate the use of any five JQuery method to manipulate HTML elements with code snippet. **(05 Marks)**
c. Write a JQuery program that performs fading in and fading out of the HTML element. **(05 Marks)**

Module-5

9 a. What is an angular JS Directive? Explain the use of the following directives with code snippet. **(08 Marks)**
b. What is the use of a filter in Angular JS? Explain the use of JSON, Currency, Number, Lowercase. **(08 Marks)**
c. Demonstrate $ scope with an example. **(04 Marks)**

OR

10 a. Write an Angular JS program to demonstrate client-side form validation. **(10 Marks)**
b. With code snippet, explain the use of controllers in Angular JS. **(10 Marks)**

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20MCA32

Third Semester MCA Degree Examination, July/August 2022
Internet of Things

Time: 3 hrs. Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
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Module-1

1 a. What is IoT? Explain in detail on genesis of IoT. **(08 Marks)**
b. What does IoT and Digitization mean? Elaborate "IoT impact on Real World". **(08 Marks)**
c. Write a Python program to count the number of words and characters in a given string. **(04 Marks)**

OR

2 a. Discuss IoT challenges. **(04 Marks)**
b. Explain the M2M IoT architecture with neat diagram. **(08 Marks)**
c. Explain the Core IoT functional stack. **(08 Marks)**

Module-2

3 a. Define Sensors and Actuators. Explain how they interact with the physical world. **(08 Marks)**
b. Define smart objects. Explain its characteristics. **(08 Marks)**
c. Explain "IoT Access Technologies. **(04 Marks)**

OR

4 a. What is SANET? Explain some advantages and disadvantages that a wireless based solution offers. **(08 Marks)**
b. Explain the LoRaWAN standardization and alliance, MAC layer. **(08 Marks)**
c. Write a python program to find area of triangle using Heron's formula without using Math library. **(04 Marks)**

Module-3

5 a. Discuss various IoT application transport methods. **(08 Marks)**
b. What is CoAP? Draw CoAP message format. Explain its all fields. **(08 Marks)**
c. Write a note on generic web-based protocols. **(04 Marks)**

OR

6 a. Describe MQTT framework and message forma in detail. **(08 Marks)**
b. Explain the key advantages of Internet Protocol. **(08 Marks)**
c. Compare between CoAP and MQTT. **(04 Marks)**

Module-4

7 a. Explain neural network in machine learning with a detailed example. **(08 Marks)**
b. Discuss the different elements of Hadoop with a neat diagram. **(08 Marks)**
c. Write a note on Massively Parallel Processing (MP) databases. **(04 Marks)**

OR

8 a. Discuss Big data analytics tools and technology. **(08 Marks)**
b. Explain the different steps and phases of OCTAVE Allegro methodology. **(08 Marks)**
c. Write a note on FAIR formal Risk Analysis. **(04 Marks)**

Module-5

9 a. Explain the different pins/parts of Arduino Uno Board. **(08 Marks)**
b. With a neat diagram, explain Raspberry Pi board. **(08 Marks)**
c. Write a Python program on Raspberry Pi to blink an LED. **(04 Marks)**

OR

10 a. Discuss the different layers of IoT Smart city Traffic architecture. **(08 Marks)**
b. With the case study, explain smart and connected parking using Raspberry Pi. **(08 Marks)**
c. What are the advantages and disadvantages of smart parking architecture? **(04 Marks)**

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20MCA32

Third Semester MCA Degree Examination, July/August 2022
Internet of Things

Time: 3 hrs. Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1 a. What is IoT? Explain in detail on genesis of IoT. **(08 Marks)**
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OR

8	a.	Discuss Big data analytics tools and technology.	(08 Marks)
	b.	Explain the different steps and phases of OCTAVE Allegro methodology.	(08 Marks)
	c.	Write a note on FAIR formal Risk Analysis.	(04 Marks)

Module-5

9	a.	Explain the different pins/parts of Arduino Uno Board.	(08 Marks)
	b.	With a neat diagram, explain Raspberry Pi board.	(08 Marks)
	c.	Write a Python program on Raspberry Pi to blink an LED.	(04 Marks)

OR

10	a.	Discuss the different layers of IoT Smart city Traffic architecture.	(08 Marks)
	b.	With the case study, explain smart and connected parking using Raspberry Pi.	(08 Marks)
	c.	What are the advantages and disadvantages of smart parking architecture?	(04 Marks)

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20MCA42

Fourth Semester MCA Degree Examination, July/August 2022
Programming using C#

Time: 3 hrs. Max. Marks:100

Note: ***Answer any FIVE full questions, choosing ONE full question from each module.***

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

Module-1

1 a. Explain the components of •Net framework with the help of architecture diagram. **(10 Marks)**
b. List and explain the different data types in C#. **(10 Marks)**

OR

2 a. Discuss between value types of reference types and write a program to explain boxing and unboxing. **(10 Marks)**
b. Write short notes on :
i) Windows work flow foundation
ii) Windows cold space and LINQ. **(10 Marks)**

Module-2

3 a. Explain partial classes and partial methods, with the help of a C# program. **(10 Marks)**
b. Write a C# program to explain accessor and mutator properties used in encapsulation. **(10 Marks)**

OR

4 a. Explain the significance of "is – a" and "has – a" relationship in inheritance with an example program. **(10 Marks)**
b. Explain the following with example :
i) Static class and static members
ii) Abstract class and abstract methods. **(10 Marks)**

Module-3

5 a. Explain the four steps involved in creating and using delegate in the program. **(10 Marks)**
b. Explain the architecture of APO•NET with a neat diagram and list the ADO•NET provided. **(10 Marks)**

OR

6 a. Write a C# program using try, catch, finally to explain any predefined exceptions. **(10 Marks)**
b. Explain the properties and methods of Data Reader and Data Adaptor class. **(10 Marks)**

Module-4

7 a. Explain various mouse events in C# windows application. **(10 Marks)**
b. Explain the WPF architecture with a neat diagram. **(10 Marks)**

OR

8 a. Explain the following :
   i) XAML elements
   ii) Makeup extension classes in XAML. **(10 Marks)**

b. Write short notes on :
   i) MDI window forms
   ii) Event –Driven GUI. **(10 Marks)**

Module-5

9 a. Explain different validation controls with suitable example supported by ASP•NET. **(10 Marks)**

b. Explain the architecture of three tier web based application. **(10 Marks)**

OR

10 a. What are Cookies? Explain session management in ASP•NET using cookies. **(10 Marks)**

b. Explain the controls from AJAX control toolkit. **(10 Marks)**

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CBCS SCHEME

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